

Project Proposal: Space-Time Travel Planner

The Space-Time Travel Planner is a project designed to solve a simple problem: many people are curious about space travel, but it is hard to imagine how it actually affects us in real life. Most people do not know exactly how long a trip to another planet takes, or how a different planet's gravity would completely change their physical weight.

The main goal of this project is to build an easy-to-use, interactive program that teaches users about the solar system in a fun way. To do this, the program will feature a simple menu where the user can pick a destination like Mars, Jupiter, or Venus. It will use math calculations to find their new weight based on the specific gravity of the planet they choose. It will also track time by capturing the current date and adding the travel days to show the exact arrival date, as well as how old the user will be when they finally land. If a user types a wrong choice, the program will handle the error safely by telling them it is an invalid choice before closing.

For the program to work, it will need a few basic inputs from the user. It will ask for their current weight in pounds, their current age in years, and their number choice for the planet destination. After processing this data, the program will output a complete "Trip Summary" that includes the destination name, the departure date, the total number of travel days, and the calculated arrival date. Finally, it will output "Planetary Metrics," which will show the user's new body weight rounded to two decimal places and their new age upon arrival rounded to one decimal place.

The program will follow a very clear step-by-step logic plan to do all of this. First, it will start by printing a welcome banner on the screen. Next, it will collect the user's weight and age, and then show the menu of planet options. The program will check the user's choice using "if/else" logic to set the correct planet name, gravity value, and travel days. After assigning these values, it will calculate the new weight by multiplying the input weight by the planet's gravity. It will also calculate the arrival date and the new age by doing simple addition and division. Finally, it will print all the summarized results on the screen and end the program.

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START
PRINT "SPACE-TIME TRAVEL PLANNER"
PRINT "weight (in lbs): "
INPUT weight
PRINT "age (in years): "
INPUT age
PRINT destination options (1: Mars, 2: Jupiter, 3: Venus)
INPUT choice

IF choice == 1 THEN
    planet = "Mars"
    gravity = 0.38
    travel_days = 212
ELSE IF choice == 2 THEN
    planet = "Jupiter"
    gravity = 2.34
    travel_days = 600
ELSE IF choice == 3 THEN
    planet = "Venus"
    gravity = 0.90
    travel_days = 150
ELSE
    PRINT "Invalid choice!"

EXIT
ENDIF

planet_weight = weight * gravity
departure_date = CURRENT_DATE
arrival_date = departure_date + travel_days
planet_age = age + (travel_days / 365)

PRINT "===== TRIP SUMMARY ====="
PRINT destination, departure_date, travel_days, arrival_date
PRINT "===== PLANETARY METRICS ====="
PRINT planet_weight (rounded to 2 decimal places)
PRINT planet_age (rounded to 1 decimal place)
PRINT "===== "
END

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